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Generative AI Notes

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1. What is Generative AI ?

Generative AI is a type of artificial intelligence that can create new content such as text, images, audio, video and code based on the patterns it has learned from large amounts of existing data.

Key points :

- Learns the underlying patterns from data.
- Generates new and original content.
- Can work with text, images, audio, video and other formats.
- Uses deep learning models, mainly large neural networks.
- Becomes better with more and high quality data.

Examples :

- Text - blog posts, summaries, chat responses.
- Image - realistic images, artwork, product designs.
- Audio - human like speech, background music.
- Video - short video clips, animations.
- Code - generate and complete code snippets.

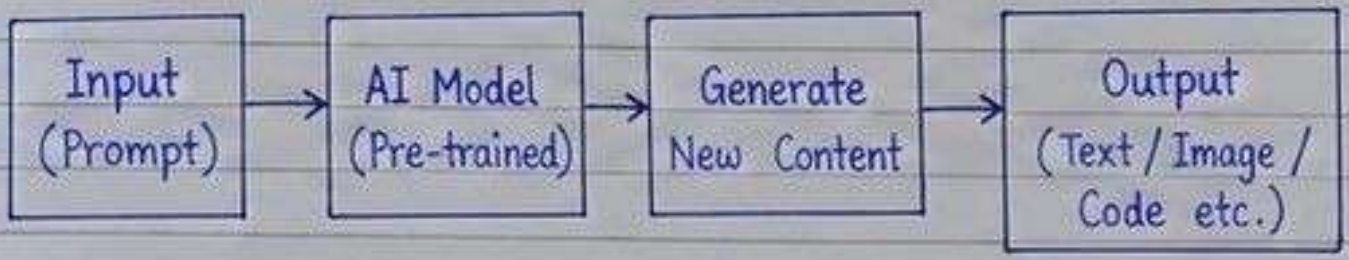
2. How Generative AI Works ?

Generative AI models learn the probability distribution of data and use this knowledge to generate new content that looks similar to the training data but is not a direct copy.

General Process :

1. Collect large amount of data.
2. Preprocess and convert data into a suitable format.
3. Train a deep learning model to learn patterns.
4. Use the trained model to generate new content based on a user input or prompt.
5. Refine the output using techniques like fine-tuning or prompt engineering.

Simple flow :



Important concept :

- The model does not understand like a human.
- It predicts the next most likely token based on the context.
- The quality of output depends on the training data, model size and prompt.

3. Types of Generative AI Models

Generative AI models can be categorised based on the type of data they generate and the architecture used. Different models are suitable for different use cases.

1. Large Language Models (LLM) - generate text.
2. Diffusion Models - generate images.
3. Generative Adversarial Networks (GAN) - generate images and videos.
4. Variational Autoencoders (VAE) - generate images, text and structured data.
5. Multimodal Models - work with multiple types of data like text, image, audio and video.

Comparison of common models :

Model Type	Data Type	Common Examples	Typical Use Cases
LLM	Text	GPT, LLaMA, Gemini	Chatbots, content generation, code
Diffusion Model	Image	Stable Diffusion, DALL·E	Image generation, design, art
GAN	Image / Video	StyleGAN	Realistic image generation
VAE	Image / Text	VAE models	Data generation, representation learning
Multimodal	Text + Image + Audio + Video	GPT-4V, Gemini	Cross modal applications

4. Large Language Models (LLM)

Large Language Models are deep learning models trained on massive amounts of text data to understand, generate and manipulate human language.

Key features :

- Trained on billions of parameters.
- Understand context and generate human like text.
- Can perform tasks like summarization, translation, question answering and code generation.
- Examples : OpenAI GPT series, Google Gemini, Meta LLaMA, Anthropic Claude.

How LLM works (Simplified) :



Applications :

- Chatbots and virtual assistants.
- Content writing and summarization.
- Code generation and debugging.
- Language translation.
- Personalized learning and tutoring.
- Business report generation.

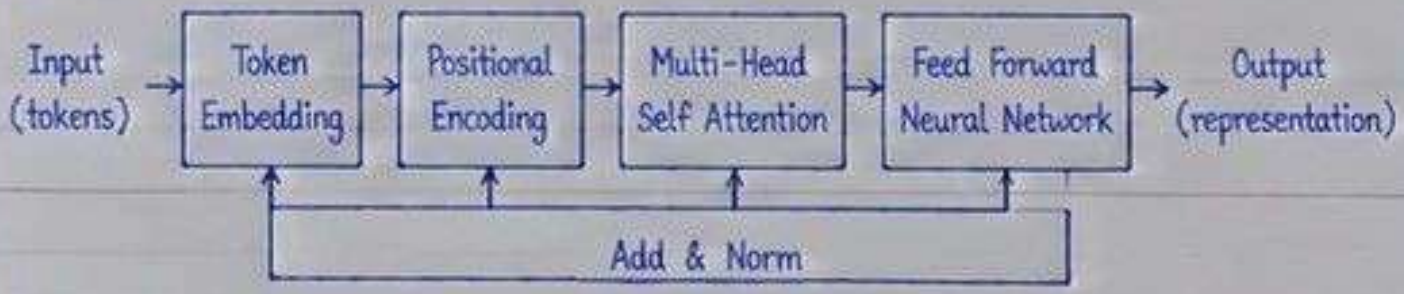
5. Transformer Architecture

Transformer is the backbone architecture used in most modern Generative AI models, especially Large Language Models. It uses the self-attention mechanism to understand the relationship between words in a sequence.

Key components :

1. Token Embedding - converts tokens into vectors.
2. Positional Encoding - adds information about the position of tokens.
3. Self Attention - helpes the model focus on important words in the input.
4. Feed Forward Neural Network - processes the output of attention layer.
5. Layer Normalization - stabilizes training.
6. Stacked Layers - multiple transformer blocks are used.

Transformer block (simplified) :



Advantages :

- Can capture long range dependencies.
- Works well with large datasets.
- Highly parallelizable.
- Scales effectively with more data and parameters.

6. Prompt Engineering

Prompt engineering is the process of designing and structuring input prompts to get the most accurate and useful response from a generative AI model.

Principles of good prompts :

1. Be clear and specific.
2. Provide context and background information.
3. Mention the desired format of the output.
4. Use step by step instructions for complex tasks.
5. Use examples if required.
6. Set the tone, style and audience.
7. Iterate and refine the prompt.

Types of prompts :

- Zero shot prompt - give a direct instruction.
- Few shot prompt - provide examples.
- Chain of thought - ask the model to think step by step.
- Role based prompt - assign a specific role.
- Constraint based prompt - set limitations.

Example :

> You are a data analyst. Explain the key reasons for the growth of electric vehicles in India. Answer in 5 bullet points with simple language for a business audience.

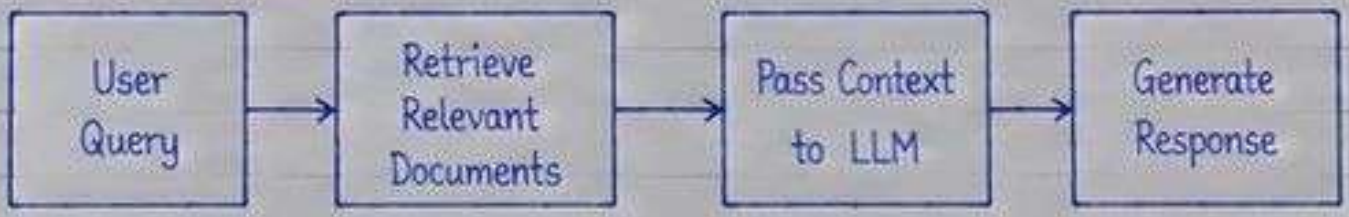
7. Retrieval Augmented Generation (RAG)

RAG is a technique that combines the knowledge retrieval system with a generative model. It fetches relevant information from external sources and then uses a language model to generate a response based on that information.

Why RAG is useful ?

- Provides up to date information.
- Reduces hallucinations.
- Can use internal company data.
- More accurate and reliable responses.

How RAG works :



Components :

1. Data Source - documents, websites, databases.
2. Embedding Model - convert text to vectors.
3. Vector Database - store and search embeddings.
4. Retriever - find relevant documents.
5. Generative Model - generate final answer.

Applications :

- Company knowledge base.
- Research and academic search.
- Customer support chatbots.
- Document summarization and Q&A systems.

8. Fine-tuning and Model Customization

Fine-tuning is the process of further training a pre-trained model on a smaller and specific dataset to adapt it to a particular task or domain.

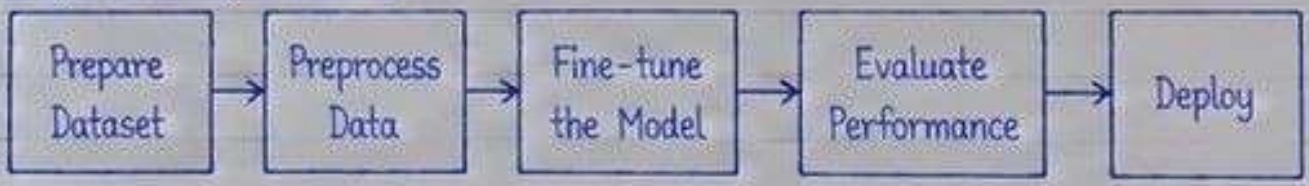
When to use fine-tuning ?

- When you have domain specific data.
- When prompt engineering is not enough.
- When you need consistent and specific output style.
- For better performance on specialised tasks.

Approaches :

1. Full Fine-tuning - update all model parameters.
2. Parameter Efficient Fine-tuning - update only a small number of parameters (e.g. LoRA, Adapters).

Steps (simplified) :



Evaluation metrics :

- Accuracy or task specific metric.
- Human evaluation.
- Response quality and relevance.
- Safety and bias check.

Use cases :

- Industry specific chatbots.
- Legal, medical or financial domain models.
- Custom writing or content generation.

9. Vector Embeddings and Similarity Search

Vector embeddings are numerical representations of text, images or other data that capture the meaning and context. They allow similar items to be found using mathematical similarity measures.

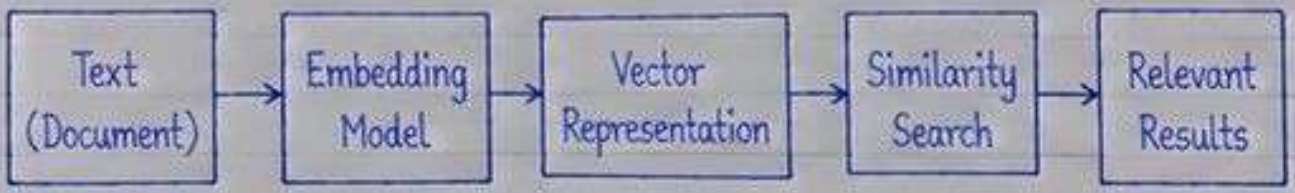
Key points :

- Convert data into high dimensional vectors.
- Similar meaning has similar vector representation.
- Used in search, recommendation and RAG systems.
- Common models : OpenAI Embeddings, Sentence Transformers.

Similarity measures :

1. Cosine Similarity - measures the angle between vectors.
2. Euclidean Distance - measures the straight line distance.
3. Dot Product - measures the similarity based on magnitude and direction.

Process flow :



Applications :

- Semantic search.
- Recommendation systems.
- Duplicate content detection.
- Personalized content delivery.
- Improved information retrieval in RAG.

10. Diffusion Models for Image Generation

Diffusion models are a class of generative models that generate high quality images by gradually removing noise from a random image. They have become very popular for image generation tasks.

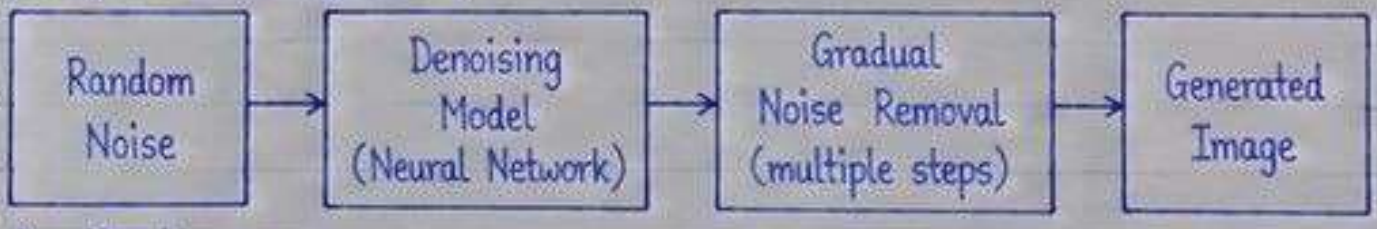
How it works :

1. Start with a random noise image.
2. Use a neural network to predict and remove noise step by step.
3. Repeat the process for many steps.
4. Final output is a high quality image.

Key features :

- Generate realistic and high resolution images.
- Can be conditioned on text (text to image).
- Examples : Stable Diffusion, DALL·E, Midjourney.
- Widely used for art, design, marketing and content creation.

Simplified diagram :



Applications :

- Text to image generation.
- Image editing and variation.
- Synthetic data generation.
- Design and creative work.
- Personalized avatars and product designs.

11. Generative Adversarial Networks (GAN)

GAN is a type of generative model that uses two neural networks, a generator and a discriminator, which compete with each other. The generator creates fake data and the discrimination tries to identify whether the data is real or fake.

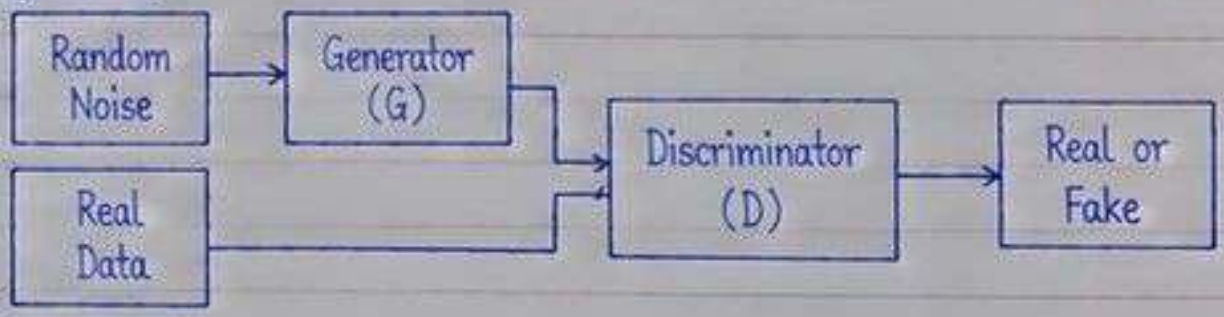
Components :

1. Generator - generates fake data (images, text etc.).
2. Discriminator - distinguishes between real and fake data.

How it works :

- Generator creates fake data.
- Discriminator checks if the data is real or fake.
- Both networks are trained together.
- Over time generator improves and creates more realistic data.

Diagram :



Applications :

- Image generation (realistic images).
- Image to image translation.
- Data augmentation.
- Style transfer.
- Used in creative design, gaming and media.

12. Variational Autoencoders (VAE)

Variational Autoencoders are generative models that learn a compressed representation of the data and then generate new data by sampling from this representation. They use an encoder network to convert the input into a probability distribution and a decoder network to generate the output from the sampled latent vector.

Key components :

1. Encoder - maps input to a latent distribution (mean and variance).
2. Latent Space - compressed representation of data.
3. Decoder - generates new data from sampled latent vector.

Key features :

- Learns a continuous latent space.
- Generates diverse and realistic outputs.
- Can interpolate between data points.
- Widely used for image, text and structured data.

Architecture (simplified) :



Applications :

- Image generation.
- Data denoising.
- Anomaly detection.
- Drug discovery and molecule generation.
- Representation learning.

13. Multimodal Generative AI

Multimodal generative AI models can understand and generate content across multiple types of data such as text, images, audio and video. These models learn joint representations and can perform tasks that involve more than one modality.

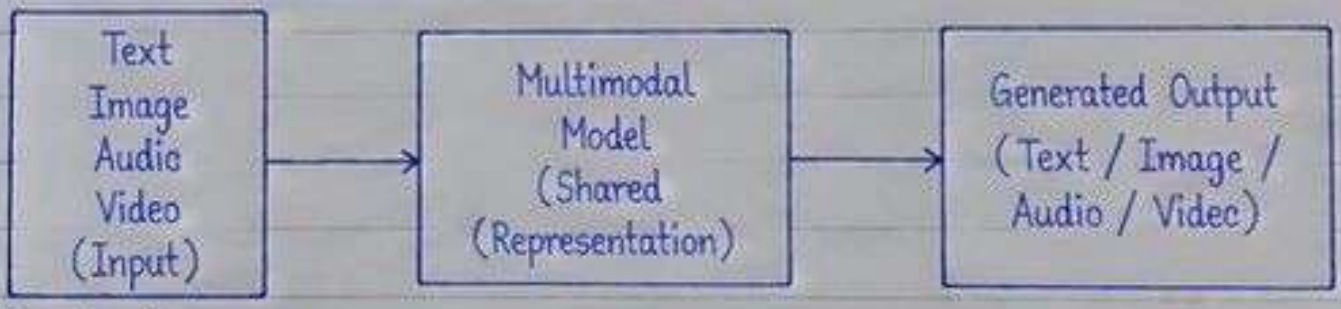
Examples of multimodal models :

- OpenAI GPT 4 with image understanding
- Google Gemini
- DALL-E with text and image
- Video generation models like Sora
- Audio and vision language models.

Key capabilities :

- Understand and generate text, images, audio and video.
- Perform cross modal reasoning.
- Create consistent outputs across different formats.
- Useful for real world applications.

How it works (simplified) :



Applications :

- Content creation.
- Virtual assistants with vision and voice.
- Video summarization.
- Education and training tools.
- Marketing and media generation.

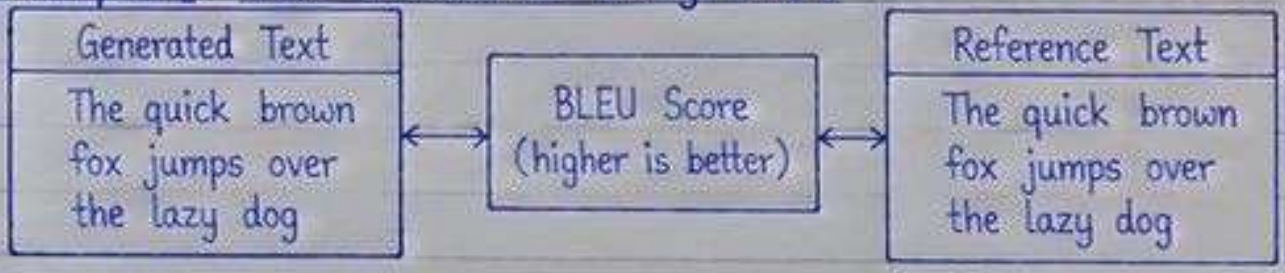
14. Evaluation Metrics for Generative Models

Evaluation of generative models is different from traditional machine learning models. Since the output is not a single fixed answer, multiple metrics are used to measure the quality, diversity and relevance of the generated content.

Common evaluation metrics :

1. BLEU - measures similarity between generated text and reference text.
2. ROUGE - used for text summarization.
3. METEOR - uses semantic matching.
4. Perplexity - measures how well the model predicts text.
5. FID (Fréchet Inception Distance) - used for image generation to compare real and generated images.
6. IS (Inception Score) - measures image quality and diversity.
7. Human Evaluation - most reliable method for assessing content quality.

Example : Text Evaluation using BLEU



Points to remember :

- No single metric is perfect.
- Use multiple metrics for better evaluation.
- Human feedback is important.
- Choice of metric depends on the type of content (text, image, audio or video).

15. Responsible AI and Safety in Generative AI

As generative AI becomes more powerful and widely used, it is important to use it responsibly. Responsible AI focuses on reducing risks, ensuring fairness and making sure the models are used in a safe and ethical way.

Key principles :

1. Fairness - avoid biased and discriminatory outputs.
2. Transparency - understand how the model works.
3. Privacy - protect user data and sensitive information.
4. Safety - prevent harmful or misleading content.
5. Accountability - take responsibility for the use of AI.

Common challenges :

- Biased or toxic outputs.
- Generation of false or misleading information.
- Privacy risks when using personal or confidential data.
- Misuse for spamming, fraud or harmful content.
- Lack of explainability in large models.

Approaches to mitigate risks :

- Use of content filters and moderation systems.
- Human review for critical applications.
- Bias detection and evaluation.
- Clear usage guidelines and policies.
- Regular monitoring and updates.

Simplified workflow :



16. Generative AI Tools and Ecosystem

There are many tools and platforms available today that make it easy to work with generative AI models. These tools provide ready to use models, APIs and development environments to build different types of applications.

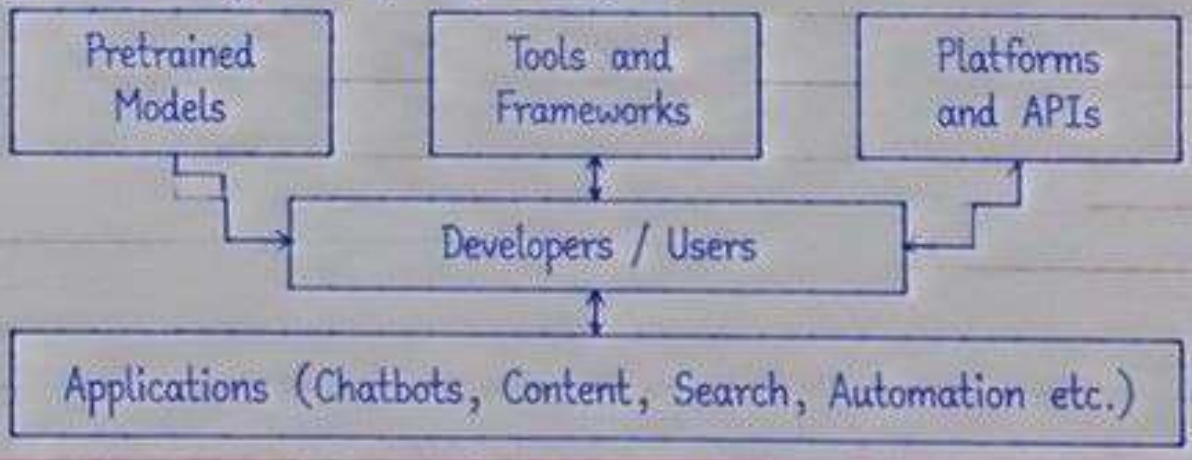
Popular tools and platforms :

1. OpenAI - GPT models, DALL-E, Whisper.
2. Google - Gemini, Imagen, Vertex AI.
3. Meta - LLaMA models.
4. Anthropic - Claude models.
5. Stability AI - Stable Diffusion.
6. Hugging Face - open source models and tools.
7. LangChain - framework to build LLM applications.
8. Microsoft - Azure OpenAI, AI services.

Types of tools :

- Model APIs - access powerful models through API.
- Development frameworks - LangChain, LlamaIndex.
- Model hosting platforms - Hugging Face, Azure, AWS.
- Application tools - ChatGPT, Copilot, Midjourney.
- No code tools - easy to build AI applications.

Ecosystem diagram (simplified) :



17. Real World Applications of Generative AI

Generative AI is being used in many industries to solve real problems and create new opportunities. It helps in automating tasks, improving productivity and creating personalized experiences.

Applications across different domains :

1. Education - personalized learning, content generation.
2. Healthcare - medical reports, drug discovery, patient support.
3. Finance - report generation, fraud detection, customer support.
4. Marketing - content creation, ad copy, campaign ideas.
5. Software Development - code generation, debugging, documentation.
6. Legal - document summarization, contract analysis.
7. HR - resume screening, personalized training.
8. E-commerce - product descriptions, recommendation content.

Some specific use cases :

- Chatbots for customer support.
- Automated report and presentation generation.
- Image and video creation for marketing.
- Code generation and automated testing.
- Virtual assistants for internal business operations.
- Personalized content and recommendations.

Benefits :

- Saves time and effort.
- Increases productivity.
- Enables creativity and innovation.
- Improves customer experience.
- Helps in data driven decision making.

18. Prompting Techniques for Better Results

Prompting is the most important skill when working with generative AI models. A well written prompt helps the model understand the task clearly and generate more accurate and useful responses.

Key elements of a good prompt :

1. Clear instruction - tell what you want.
2. Context - provide background information.
3. Input data - give relevant data if needed.
4. Output format - mention the desired format.
5. Constraints - set limitations or conditions.

Common prompting techniques :

- Zero shot prompting - direct instruction without examples.
- Few shot prompting - provide examples in the prompt.
- Chain of thought prompting - ask the model to think step by step.
- Role based prompting - assign a specific role to the model.
- Instruction tuning - give detailed and structured instructions.

Example (role based prompt) :

> You are a career counselor. Suggest a 6 month learning plan for a beginner who wants to learn generative AI. Provide the plan in a table format with weekly topics and resources.

Benefits :

- More accurate and relevant responses.
- Better control over the output.
- Useful for complex and domain specific tasks.
- Improves consistency in results.

19. RAG vs Fine-tuning vs Prompt Engineering

There are different approaches to improve the performance of generative AI models. The three most common methods are prompt engineering, retrieval augmented generation (RAG) and fine-tuning. Each method has its own use cases, advantages and limitations.

Feature	Prompt Engineering	RAG	Fine-tuning
What it does	Better prompts for existing model	Uses external data for response	Trains the model on custom data
Data required	No additional data	External documents	Labeled training data
Model update	No change	No change	Model parameters are updated
Cost	Low	Medium	High
Use cases	General questions, simple tasks	Domain specific knowledge, up to date information	Specialized tasks, custom behaviour
Example	Write a summary of this article	Answer using company documents	Legal, medical or industry specific model

When to use which method ?

- Use prompt engineering when you need quick and simple solutions.
- Use RAG when you need accurate and up to date information from external sources.
- Use fine-tuning when you have a large dataset and need domain specific or specialized behaviour.
- In many real world applications, a combination of these methods is also used.